

MA 562: An Introduction to Bayesian Analysis

Course Code: MA 562

Course Name: An Introduction to Bayesian Analysis

L-T-P-C: 3-0-2-4

Prerequisites: Probability and Statistics (MA 524) and/or IC 252

Intended for: B.Tech 3rd and 4th year/ MS/MSc/PhD

Mutual Exclusion: NA

Elective or Core: Elective

Approval: 61st BoA

Preamble

Bayesian Analysis provides a principled framework for statistical inference under uncertainty, treating unknown quantities as random variables and updating beliefs using observed data. This course offers a foundational introduction to Bayesian thinking, modeling, and computation, with an emphasis on clarity, interpretability, and practical data analysis. Unlike advanced Bayesian courses that focus on research-level modeling or project-based exploration, this course is designed to build strong conceptual and computational foundations. The course progresses gradually from the philosophical and probabilistic basis of Bayesian inference to standard Bayesian models and basic computational tools. The emphasis is on understanding core ideas, posterior reasoning, and implementation using simple computational techniques, without delving into advanced or specialized models. Overall, the course equips students with a basic of Bayesian inference, preparing them for advanced research and practical applications in statistics, machine learning, and related fields.

Course Modules

- **Bayesian Foundations and Probability Review:** Review of probability concepts relevant to Bayesian inference, conditional probability and Bayes' theorem, Bayesian versus frequentist paradigms, likelihood principle and subjective probability, Simple motivating examples. (6 Lectures)
- **Bayesian Inference and Prior Specification:** Bayesian inference framework, prior, likelihood, and posterior distributions, types of priors: informative, weakly informative, and non-informative priors, Conjugate priors and analytical posteriors, posterior summaries: mean, median, credible intervals. (6 Lectures)
- **Bayesian Models for Standard Data Types:** Bayesian inference for discrete data (Binomial, Poisson models), Bayesian inference for continuous data (Normal model), Predictive distributions, posterior predictive checks (basic ideas), interpretation of Bayesian results. (6 Lectures)
- **Bayesian Computation and Simulation:** Motivation for Bayesian computation, Deterministic approximation methods (basic overview), introduction to Monte Carlo methods, Markov chain Monte Carlo (MCMC), Metropolis–Hastings algorithm, Assessing convergence and Monte Carlo error. (8 Lectures)
- **Bayesian Linear Regression Models and Model Comparison:** Bayesian multiple linear regression, Random effects and shrinkage interpretation, Prior choices for regression parameters, Model comparison and model selection, Bayes factors and information criteria (introductory level). (6 Lectures)
- **Approximate Bayesian Methods and Hierarchical Modeling:** Motivation for hierarchical modeling in complex data, Motivation for approximate Bayesian inference, Approximate inference for hierarchical models, Laplace approximation and normal approximations to the posterior, variational Bayes: basic ideas

and mean-field approximation, expectation propagation (conceptual overview), approximate Bayesian computation (ABC): intuition and simple algorithms, Strengths and limitations of approximate Bayesian methods. (10 Lectures)

Lab Class

The laboratory sessions focus on implementing Bayesian methods using statistical computing tools such as R and basic Bayesian software. Students will compute posterior distributions, generate posterior samples, summarize uncertainty, and analyze real and simulated datasets corresponding to the lecture material.

Project and Case Studies

This course will give a certain weight to project and case studies, implementing Bayesian hierarchical models using MCMC and approximate methods in day-to-day real-life problems.

Text Books

1. Gelman, A., Carlin, J., Stern, H., Dunson, D., Vehtari, A., & Rubin, D. *Bayesian Data Analysis*. Chapman & Hall/CRC (2013).
2. Robert, C. P., & Casella, G. *Monte Carlo Statistical Methods*. Springer (2004).

Reference

1. Marin, J.-M., & Robert, C. P. *Bayesian Essentials with R*. Springer (2014).
2. Ghosh, J. K., Delampady, M., & Samanta, T. *An Introduction to Bayesian Analysis*. Springer (2006).